

B. Tech Degree IV Semester Examination April 2011**ME 404 APPLIED THERMODYNAMICS***(2006 Scheme)*

Time : 3 Hours

Maximum Marks : 100

PART – A(Answer ALL questions)

(8 x 5 = 40)

- I. (a) What do you understand by perpetual motion machine of first kind?
 (b) What is Helmholtz function? Give its significance.
 (c) Draw Mollier Diagram and show various property lines.
 (d) Explain the terms –
 (i) Absolute humidity
 (ii) Relative humidity
 (iii) Dry Bulb temperature
 (iv) Wet Bulb temperature.
 (e) Write a short note on Rotary Compressor.
 (f) Derive an expression for the average gas constant for a mixture of three gases.
 (g) Define Colorific value and methods to determine it.
 (h) Define Adiabatic Flame temperature. What influence does excess air have on its value?

PART – B

(4 x 15 = 60)

- II. (a) Derive Clavius Clayperon Equation. (10)
 (b) Prove that adiabatic mixing of two fluids is irreversible. (5)

OR

- III. (a) A reversible heat engine receives heat from two constant temperature sources at 870°K and 580°K and rejects 50KJ/sec to a sink of 290°K . If the engine output is 85KW , calculate heat supplied by each source and efficiency of engine. (10)
 (b) Explain Increase of Entropy principle. (5)

- IV. (a) A pressure cooker contains 1.5Kg of steam at 5 bar and 0.9 dryness when the gas was turned off. Determine the quantity of heat rejected by pressure cooker when the pressure in the cooker falls to 1 bar . (10)
 (b) Draw T-S diagram for Water and explain steam generation process in it. (5)

OR

- V. (a) Briefly describe the Modified Rankine Cycle and derive its efficiency. (10)
 (b) Write a short note on Modern Steam generator. (5)

- VI. (a) Two vessels A and B each of volume 3m^3 may be connected by a tube of negligible volume. Vessel A contains air at 0.7MPa , 95°C while vessel B contains air at 0.35MPa , 205°C . Find the change in entropy when air in A mixes with B adiabatically and completely. For air $C_p=1\text{kJ/Kg.K}$ and $R = 0.287\text{ kJ/Kg.K}$. (10)
 (b) Write a note of Molecular weight and gas constant of mixture of gases. (5)

OR

- VII. (a) A 0.5m^3 tank contains air at a pressure of 7Mpa and 250°C and is perfectly insulated from surroundings. A valve on the tank is opened and air is discharged until the pressure drops to 400KPa . Calculate the mass of air discharged from the tank. (10)
 (b) Write a Note on Dalton's law of Partial pressures. (5)

- VIII. (a) With a neat sketch describe a Junkers Gas Calorimeter. (10)
 (b) Define the meaning of excess air and what is its significance in combustion process. (5)

OR

- IX. (a) A fuel has the following composition by mass carbon 86% , Hydrogen 11.75% , Oxygen 2.25% . Calculate the theoretical air supply per Kg of fuel and mass of products of combustion per Kg of fuel. (10)
 (b) Distinguish between Higher and Lower Calorific value of a fuel. (5)
